Sri Lanka Institute of Information Technology



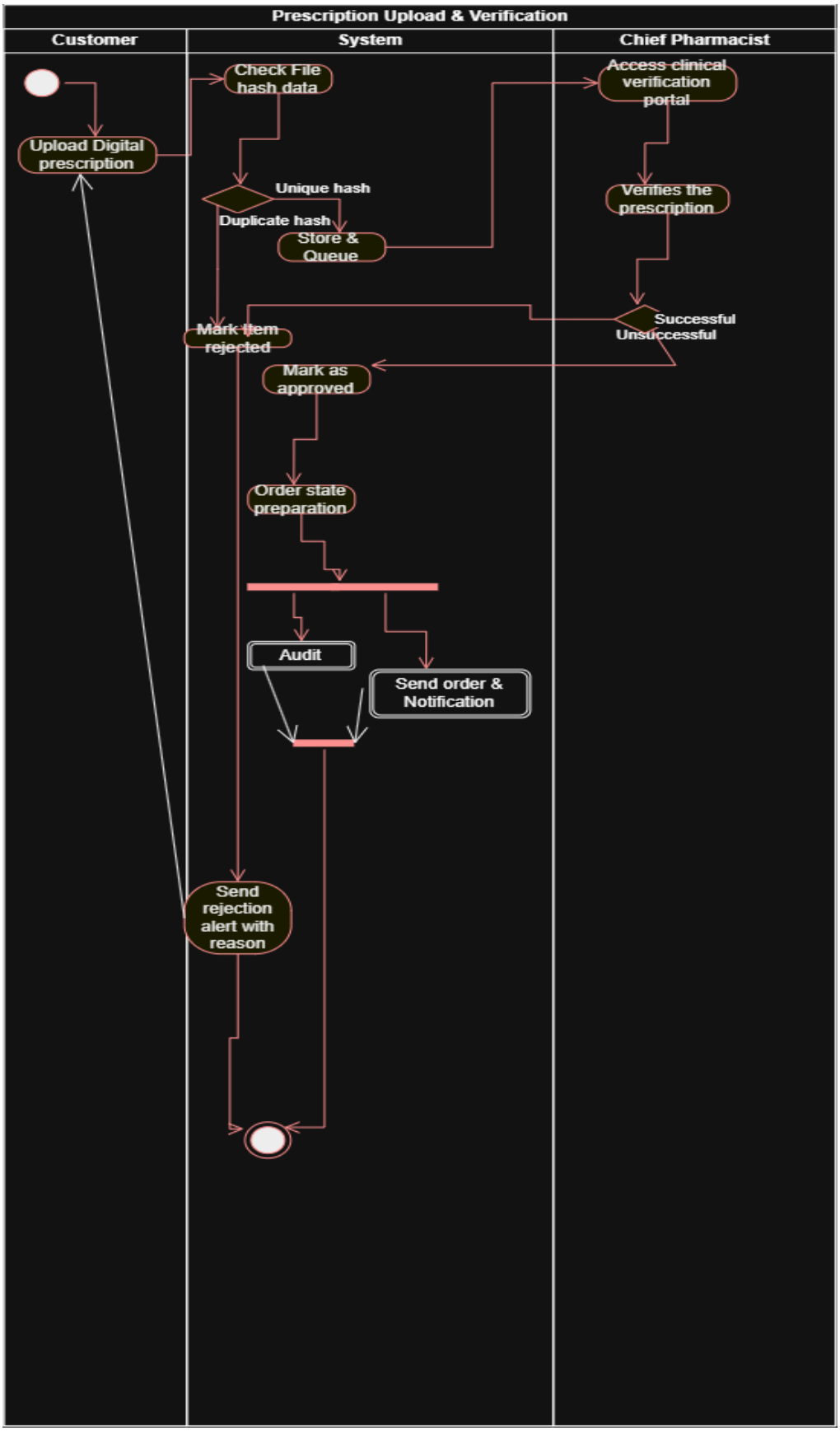
SE2030

Software Engineering

Web-based Medicine Ordering System

Activity Diagram

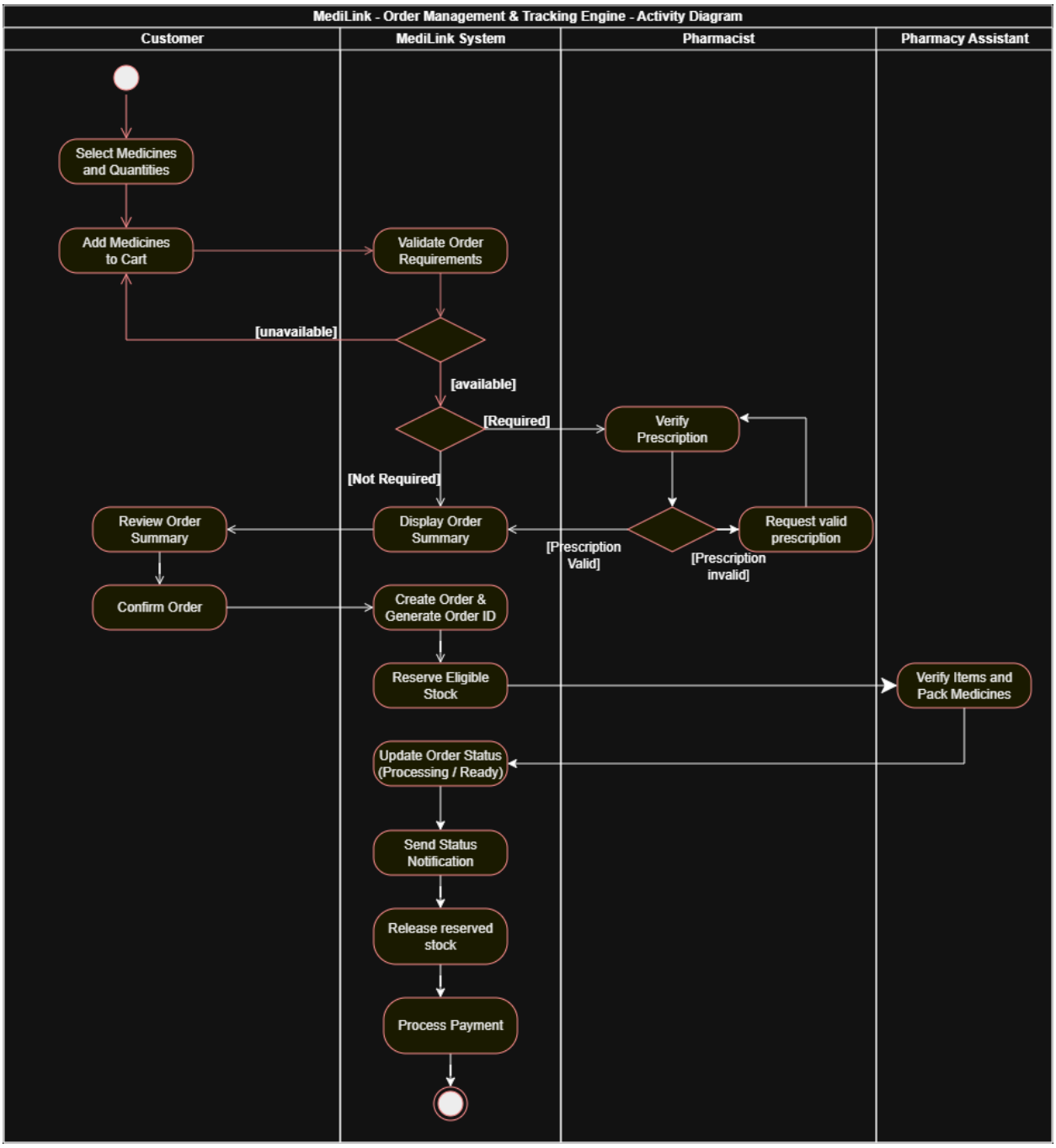
Project Group - 2026-Y2-S1-MLB-B3G1-08	
Group members	
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Prescription Upload & Verification
Customer
System
Chief Pharmacist
Check File hash data
Access clinical verification portal
Upload Digital prescription
Unique hash
Duplicate hash
Store & Queue
Verifies the prescription
Mark item rejected
Successful
Unsuccessful
Mark as approved
Order state preparation
Audit
Send order & Notification
Send rejection alert with reason

Use Case 01: Prescription Upload & Clinical Verification Engine

Number	UC-01
Name	Upload and Verify Clinical Prescription
Summary	A Patient uploads a digital prescription for restricted medication, and a Licensed Pharmacist reviews, validates, and approves or rejects it.
Priority	5 (Highest Priority)
Preconditions	1. Patient is registered and authenticated in the system. 2. Patient has added prescription-required drugs to their shopping cart.
Postconditions	1. Prescription record and file hash are securely stored. 2. Order state transitions to 'Prescription Approved' or 'Prescription Rejected'. 3. Immutable audit log entry of clinical decision is recorded.
Primary Actor(s)	Patient (Customer)
Secondary Actor(s)	Licensed Pharmacist, System Admin (Audit & Hashing)
Trigger	Patient proceeds to checkout with restricted medicines and initiates prescription upload.
Main Scenario	**Action (System-User Workflow)**
Step 1 (User)	Patient selects and uploads an image/PDF file of the doctor's prescription.
Step 2 (System)	Calculates SHA-256 hash to prevent duplicate submissions, stores file in secure storage, and adds order to 'Pending Review' queue.
Step 3 (User)	Licensed Pharmacist accesses the verification portal and opens the prescription inspector.
Step 4 (System)	Displays high-resolution image alongside patient clinical history and requested drug dosages.
Step 5 (User)	Pharmacist verifies SLMC registration number, validity dates, and dosage, then clicks 'Approve'.
Step 6 (System)	Updates status to 'Approved', advances order workflow, logs audit timestamp, and sends confirmation notification.

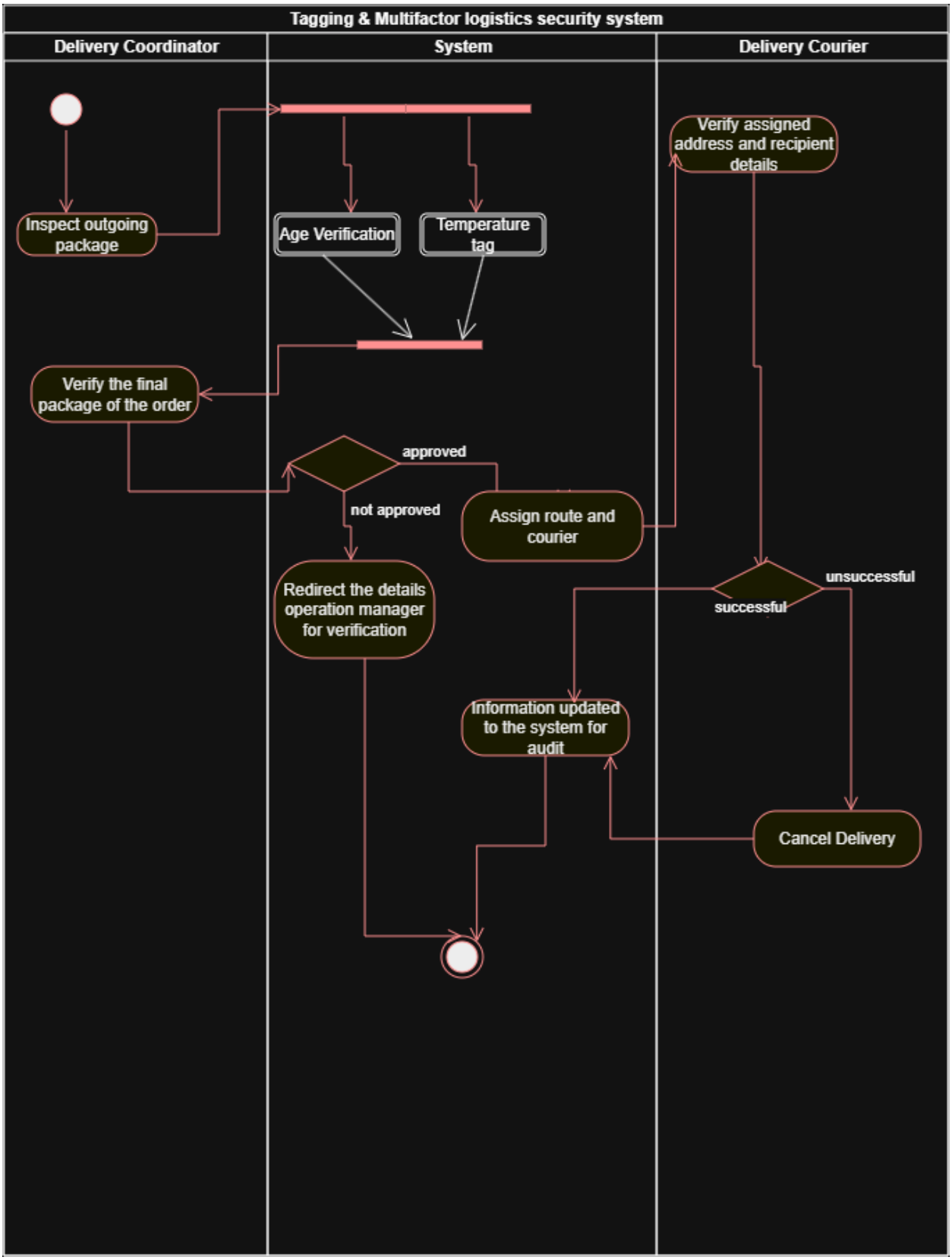
Extensions	Branching Action / Alternative Path
Step 2a	Duplicate Hash Detected: System detects identical hash in database -> Aborts upload and alerts patient that prescription was previously used.
Step 5a	Prescription Rejected: Pharmacist selects rejection reason -> System marks order item rejected, cancels item, and sends alert to patient.
Open Issues	**Resolution Notes**
Issue 1	Define fallback procedures if the pending clinical verification queue exceeds maximum SLA turnaround times during peak hours.


MediLink - Order Management & Tracking Engine - Activity Diagram
Customer
MediLink System
Pharmacist
Pharmacy Assistant
Select Medicines and Quantities
Add Medicines to Cart
Validate Order Requirements
[unavailable]
[available]
[Required]
Verify Prescription
[Not Required]
Review Order Summary
Display Order Summary
[Prescription Valid]
[Prescription invalid]
Request valid prescription
Confirm Order
Create Order & Generate Order ID
Reserve Eligible Stock
Verify Items and Pack Medicines
Update Order Status (Processing / Ready)
Send Status Notification
Release reserved stock
Process Payment

Use Case 02: Smart Inventory, Batch & FEFO Expiry Management System

Number	UC-02
Name	Allocate Stock Using First-Expired-First-Out (FEFO) Strategy
Summary	A Pharmacy Assistant fulfills order items while the system automatically designates and allocates stock from the earliest expiring available batch.
Priority	5 (Highest Priority)
Preconditions	1. Customer order is approved and pending physical packing. 2. Active batches with manufacture/expiry dates and quantities exist in inventory.
Postconditions	1. Stock deduction is executed against the specific earliest-expiring batch. 2. Quarantined or expired batches remain strictly locked from allocation.
Primary Actor(s)	Pharmacy Assistant
Secondary Actor(s)	Operations Manager (Threshold Alerts)
Trigger	Pharmacy Assistant opens the fulfillment packing dashboard and selects an approved order.
Main Scenario	**Action (System-User Workflow)**
Step 1 (User)	Pharmacy Assistant selects an approved order from the order packing queue.
Step 2 (System)	Executes FEFO query: filters valid batches by nearest expiration date and reserves required quantities.
Step 3 (System)	Generates digital pick list specifying exact Batch Number, shelf location, and quantity.
Step 4 (User)	Pharmacy Assistant picks the medicine from the shelf and scans the package barcode.
Step 5 (System)	Verifies scanned barcode matches the allocated FEFO batch and executes inventory deduction.
Step 6 (User)	Pharmacy Assistant confirms item packaging and marks task complete.
Extensions	**Branching Action / Alternative Path**

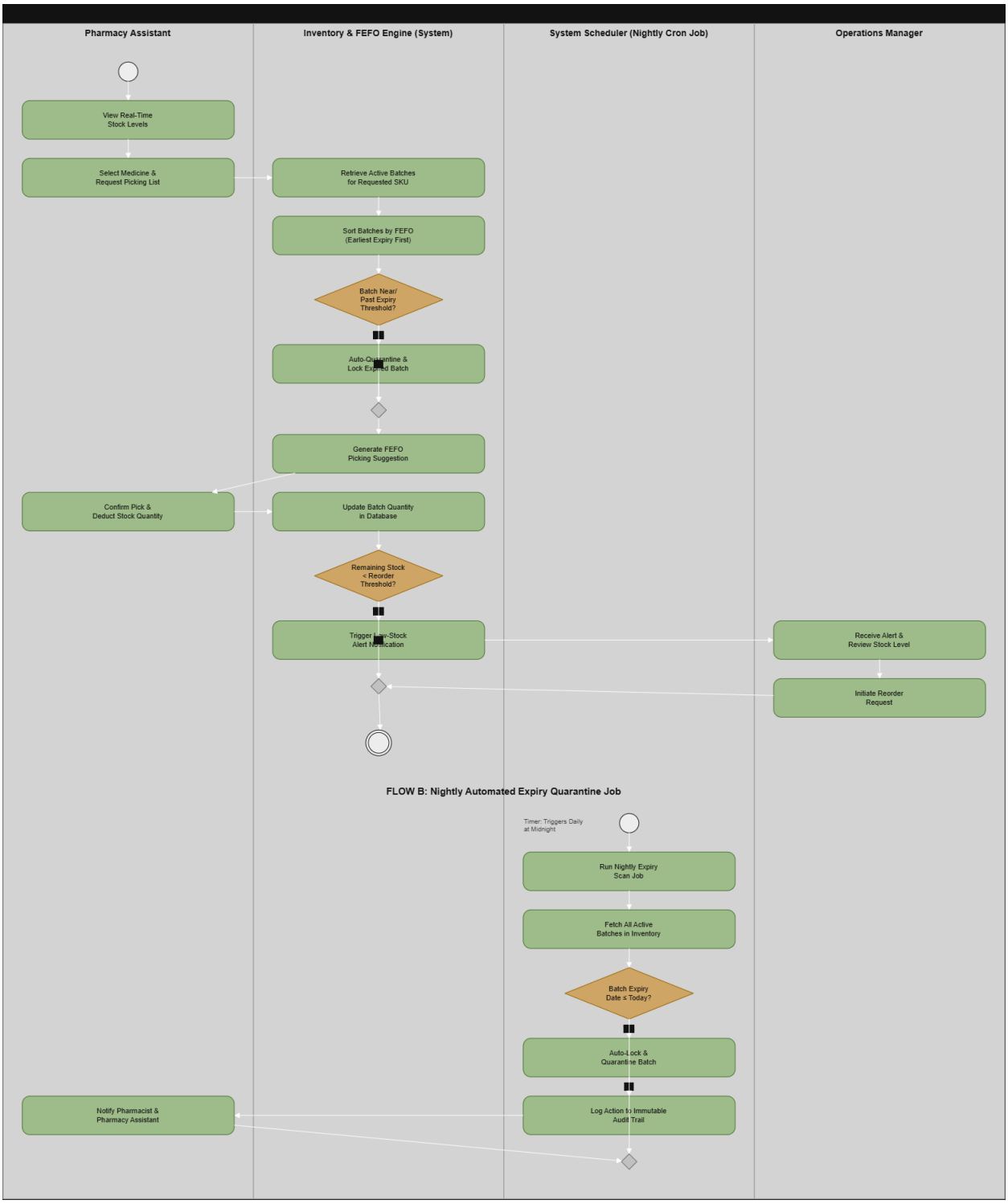
Step 2a	Expired Batch Detected: System automatically locks expired batch, flags it for quarantine, and picks next available valid batch.
Step 4a	Mismatched Barcode Scan: Assistant scans incorrect batch -> System alerts assistant with audio-visual error modal.
Step 5a	Stock Below Reorder Level: Stock deduction trips minimum threshold -> System sends automated alert to Operations Manager.
Open Issues	**Resolution Notes**
Issue 1	Handling workflow when single-batch stock quantity is insufficient and requires multi-batch split allocation.


Tagging & Multifactor logistics security system
Delivery Coordinator
System
Delivery Courier
Inspect outgoing package
Age Verification
Temperature tag
Verify assigned address and recipient details
Verify the final package of the order
approved
not approved
Assign route and courier
Redirect the details operation manager for verification
unsuccessful
successful
Information updated to the system for audit
Cancel Delivery

Use Case 04: Cold-Chain Tagging & Multi-Factor Logistics Security System

Number	UC-04
Name	Enforce Cold-Chain Handling & Age-Verification Security Protocols
Summary	A Delivery Coordinator configures and monitors mandatory cold-chain compliance and recipient age-verification protocols for sensitive and controlled medications.
Priority	5 (Highest Priority)
Preconditions	1. Items are flagged as temperature-sensitive (2°C–8°C) or age-restricted in catalog. 2. Order package is sealed and staged in dispatch bay.
Postconditions	1. Courier manifest includes strict temperature handling checklists. 2. Age-verification barrier is enforced on delivery terminal prior to release.
Primary Actor(s)	Delivery Coordinator
Secondary Actor(s)	Delivery Courier, Pharmacist
Trigger	Order containing cold-chain or controlled medication enters dispatch preparation.
Main Scenario	**Action (System-User Workflow)**
Step 1 (User)	Delivery Coordinator inspects outgoing package manifest in dispatch portal.
Step 2 (System)	Applies 'Cold-Chain 2°C–8°C' tag and enforces 'requires_age_check' verification flag.
Step 3 (User)	Delivery Coordinator verifies insulated cooler packaging and assigns route to courier.
Step 4 (System)	Pushes handling requirements and temperature inspection checklist to courier mobile app.
Step 5 (User)	Delivery Courier inspects recipient government ID at doorstep to verify legal age.
Step 6 (System)	Logs age verification and temperature sign-off into logistics audit log.

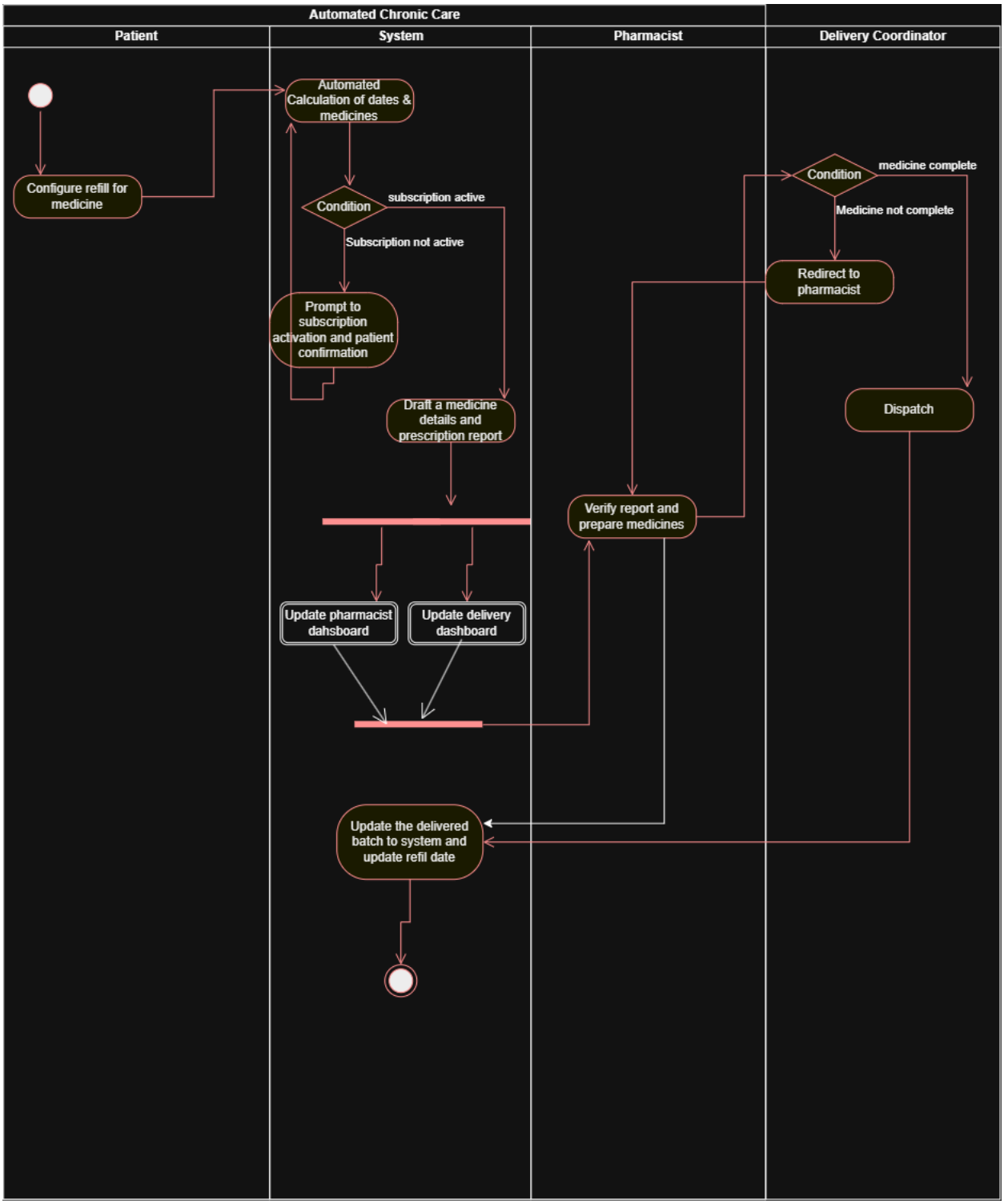
Extensions	Branching Action / Alternative Path
Step 3a	Temperature Excursion Breach: Carrier temperature exceeds 8°C prior to departure -> System halts dispatch and triggers pharmacist alert.
Step 5a	Underage Recipient / Invalid ID: Recipient cannot produce proof of age -> Courier marks verification failed and returns package to pharmacy.
Open Issues	**Resolution Notes**
Issue 1	Standardizing recipient ID photo capture rules in compliance with local personal data protection laws.


Pharmacy Assistant
Inventory & FEFO Engine (System)
System Scheduler (Nightly Cron Job)
Operations Manager
View Real-Time Stock Levels
Select Medicine & Request Picking List
Retrieve Active Batches for Requested SKU
Sort Batches by FEFO (Earliest Expiry First)
Batch Near/ Past Expiry Threshold?
Auto-Quarantine & Lock Expired Batch
Generate FEFO Picking Suggestion
Confirm Pick & Deduct Stock Quantity
Update Batch Quantity in Database
Remaining Stock < Reorder Threshold?
Trigger Low-Stock Alert Notification
Receive Alert & Review Stock Level
Initiate Reorder Request
FLOW B: Nightly Automated Expiry Quarantine Job
Timer: Triggers Daily at Midnight
Run Nightly Expiry Scan Job
Fetch All Active Batches in Inventory
Batch Expiry Date ≤ Today?
Auto-Lock & Quarantine Batch
Log Action to Immutable Audit Trail
Notify Pharmacist & Pharmacy Assistant

Use Case 03: Real-Time Order Processing & State Workflow Tracking

Number	UC-03
Name	Process Real-Time Order State Transitions & Customer Tracking
Summary	The Operations Manager oversees fulfillment pipelines while the system executes state-machine transitions and pushes real-time WebSocket progress updates to the Customer.
Priority	5 (Highest Priority)
Preconditions	1. Customer has successfully placed an order. 2. Backend WebSocket server and state event listeners are operational.
Postconditions	1. Order state advances sequentially (Placed -> Verified -> Packed -> Dispatched -> Delivered). 2. Real-time visual progress stepper updates dynamically on customer device.
Primary Actor(s)	Operations Manager
Secondary Actor(s)	Customer
Trigger	Operations Manager or operational staff update the processing milestone of an active order.
Main Scenario	**Action (System-User Workflow)**
Step 1 (User)	Operations Manager monitors the active order fulfillment pipeline dashboard.
Step 2 (User)	Internal pharmacy staff mark order status as 'Packed' and staged for courier handover.
Step 3 (System)	Order state engine validates transition, updates database record, and logs timestamp.
Step 4 (System)	Publishes real-time status event across WebSocket gateway to customer application.
Step 5 (User)	Customer observes live UI progress stepper move to 'Packed' without refreshing page.

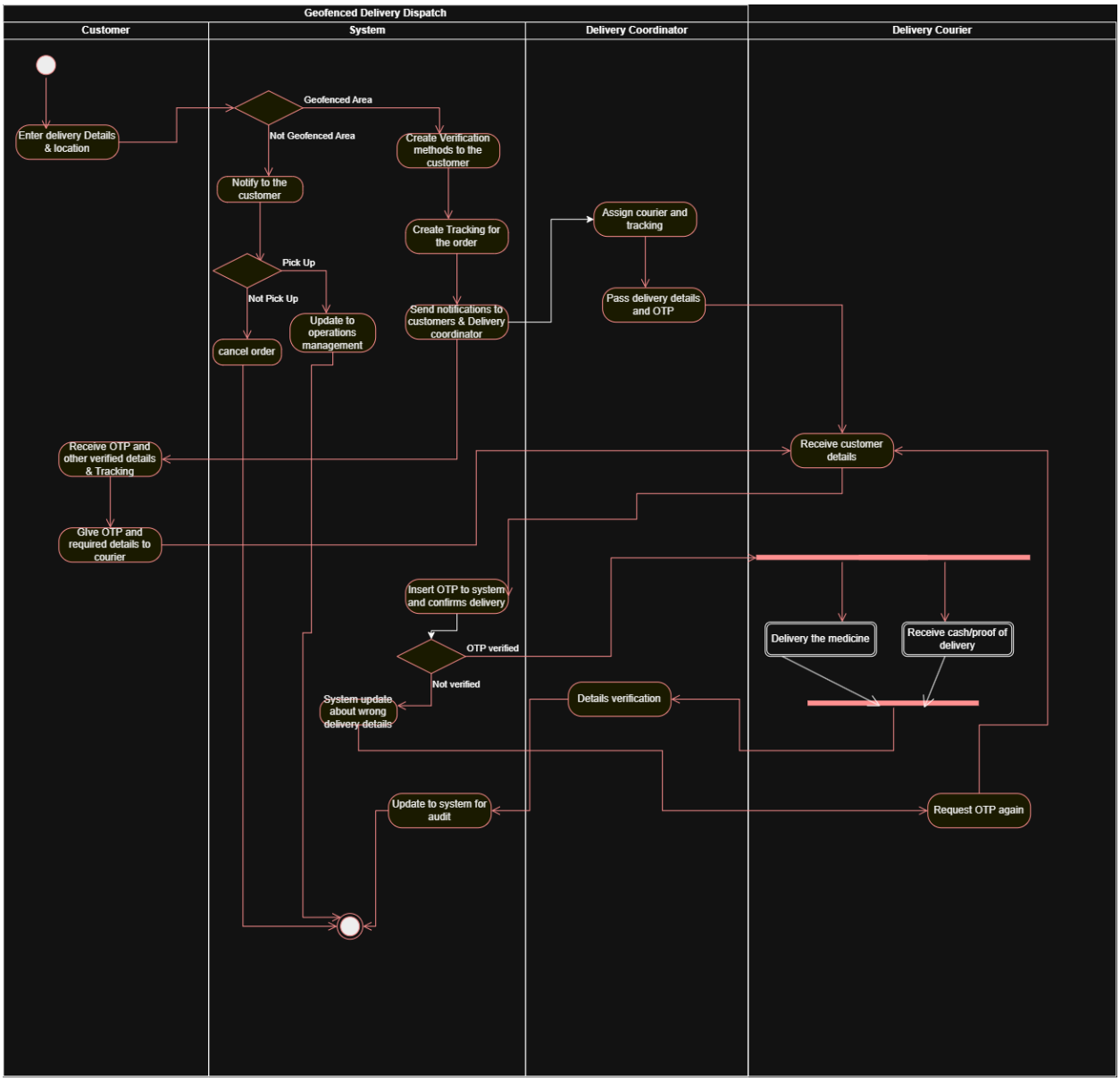
Step 6 (System)	Recalculates operational turnaround time (TAT) metrics on Operations dashboard.
Extensions	**Branching Action / Alternative Path**
Step 2a	Customer Cancellation: Customer cancels order before 'Packed' state -> System halts workflow, restores inventory, and logs cancellation.
Step 2b	Late Cancellation Request: Customer requests cancellation after 'Packed' state -> System rejects cancellation and prompts customer service flow.
Open Issues	**Resolution Notes**
Issue 1	Establishing automatic WebSocket reconnect policies during mobile client data drops or network switching.


Automated Chronic Care
Patient
System
Pharmacist
Delivery Coordinator
Configure refill for medicine
Automated Calculation of dates & medicines
Condition
subscription active
Subscription not active
Prompt to subscription activation and patient confirmation
Draft a medicine details and prescription report
Update pharmacist dahsboard
Update delivery dashboard
Verify report and prepare medicines
Condition
medicine complete
Medicine not complete
Redirect to pharmacist
Dispatch
Update the delivered batch to system and update refil date

Use Case 05: Automated Chronic Care Refill & Subscription Management Engine

Number	UC-05
Name	Manage Recurring Chronic Care Medication Refill Subscription
Summary	A Patient sets up and manages an automated monthly recurring prescription refill schedule for chronic health conditions (e.g., diabetes, hypertension).
Priority	3 (Medium Priority)
Preconditions	1. Patient holds a verified long-term prescription with allowable refill allocations. 2. Valid payment method or COD preference is linked to patient profile.
Postconditions	1. Recurring refill subscription schedule is stored in database. 2. System background worker generates draft refill orders 3 days prior to due dates.
Primary Actor(s)	Patient (Chronic Care)
Secondary Actor(s)	Licensed Pharmacist (Renewal Reviewer)
Trigger	Patient navigates to Chronic Care tab and initiates subscription setup.
Main Scenario	**Action (System-User Workflow)**
Step 1 (User)	Patient selects prescribed chronic medicine, sets 30-day recurring interval, and enters address.
Step 2 (System)	Validates prescription eligibility and stores automated recurring schedule rules.
Step 3 (System)	Runs automated cron worker 3 days before due date, creates draft order, and sends reminder.
Step 4 (User)	Patient confirms upcoming recurring dispatch via SMS / portal reminder.
Step 5 (System)	Routes draft order to Pharmacist queue for refill quota verification.
Step 6 (User)	Pharmacist confirms remaining refill balance and authorizes auto-order dispatch.

Extensions	Branching Action / Alternative Path
Step 1a	Patient Pauses Refill: Patient toggles subscription pause while traveling -> System halts auto-generation until resumed.
Step 2a	Refill Quota Expired: Prescription has 0 remaining refills -> System alerts patient to book doctor consultation and upload new prescription.
Open Issues	**Resolution Notes**
Issue 1	Handling automated retry strategies if recurring card pre-authorization fails during cron execution.


Geofenced Delivery Dispatch
Customer
System
Delivery Coordinator
Delivery Courier
Enter delivery Details & location
Geofenced Area
Not Geofenced Area
Create Verification methods to the customer
Notify to the customer
Create Tracking for the order
Assign courier and tracking
Pick Up
Not Pick Up
Update to operations management
cancel order
Send notifications to customers & Delivery coordinator
Pass delivery details and OTP
Receive OTP and other verified details & Tracking
Receive customer details
Give OTP and required details to courier
Insert OTP to system and confirms delivery
Delivery the medicine
Receive cash/proof of delivery
OTP verified
Not verified
System update about wrong delivery details
Details verification
Update to system for audit
Request OTP again

Use Case 06: Geofenced Delivery Dispatch, OTP Verification & Multi-Channel Notification

Number	UC-06
Name	Complete Geofenced Dispatch and OTP Secure Handover
Summary	A Delivery Courier uses geofenced routing tools to navigate to the customer and completes medication handover using a secure One-Time Password (OTP) verification code.
Priority	3 (Medium Priority)
Preconditions	1. Customer address is verified within active delivery geofence polygons. 2. Order is out for delivery with encrypted OTP generated in backend.
Postconditions	1. Order status is marked as 'Delivered' with GPS coordinates and timestamp. 2. Finalized transactional receipt is automatically emailed to customer.
Primary Actor(s)	Delivery Courier
Secondary Actor(s)	Customer, SMS/Email Notification Gateway
Trigger	Delivery Courier arrives at customer destination and requests handover verification.
Main Scenario	**Action (System-User Workflow)**
Step 1 (User)	Delivery Courier navigates to customer location using integrated in-app GPS map.
Step 2 (System)	Detects geofence arrival and automatically dispatches 6-digit OTP to customer phone/email.
Step 3 (User)	Customer provides the received 6-digit OTP code to the courier at delivery.
Step 4 (User)	Delivery Courier enters OTP into mobile terminal app and submits for validation.
Step 5 (System)	Backend verifies OTP match against encrypted record and captures timestamp & GPS.
Step 6 (System)	Marks order as 'Delivered' and triggers automated PDF receipt to customer email.

Extensions	Branching Action / Alternative Path
Step 1a	Out of Geofence Checkout: Address outside active boundary -> Checkout is blocked with out-of-coverage advisory.
Step 4a	Incorrect OTP Entered: Courier enters invalid OTP -> System rejects submission and locks after 3 failed attempts.
Open Issues	**Resolution Notes**
Issue 1	Offline OTP fallback protocol when delivery takes place in areas with zero cellular reception.

Student ID	Student Name	Assigned Core Function
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IT25103946	Vinogikan M.	Function 03: Real-Time Order Processing & State Workflow Tracking
IT25101882	Widyawansa T.D.	Function 04: Cold-Chain Medication Tagging & Multi-Factor Logistics Security
IT25103725	Hathoon M.F.S.	Function 05: Automated Chronic Care Refill & Subscription Management Engine
IT25100979	Kookulan E.	Function 06: Geofenced Delivery Dispatch, OTP Handover & Multi-Channel Notification Engine